

Database Normalization Progress & Next Steps

☒ Completed Today (Database Cleanup Phase)

Tables Removed

- **bls_data_versions** - Successfully dropped (backup saved)
 - Had 1 row of metadata
 - Schema preserved in backup file
 - Table was not used in production API

Other Target Tables

The following tables were already cleaned up in previous sessions:

- **major_groups** ✖ (already removed)
- **occupation_categories** ✖ (already removed)
- **occupations_normalized** ✖ (already removed)
- **occupations_old** ✖ (already removed)
- **occupations_test** ✖ (already removed)

Backup & Documentation

- ☒ Created **scripts/cleanup-database.js** - automated cleanup script
- ☒ Generated **scripts/database-cleanup-backup.json** - backup metadata
- ☒ Verified current database state (17 tables remaining)

Current Database State

Tables Currently In Use (Keep & Monitor)

1. **occupations** - Main table used by rate calculator API
2. **seo_intake_forms** - SEO intake form submissions
3. **sqlite_sequence** - SQLite internal auto-increment tracking

Tables Under Review (Next Phase)

4. **occupation_data** - Needs structure/usage analysis
5. **projections** - Needs structure/usage analysis
6. **bls_special_tables** - Needs structure/usage analysis

BLS Data Tables (Normalization Candidates)

7-17. **bls_table_1_*** series:

- **bls_table_1_1**, **bls_table_1_2**, **bls_table_1_3**, **bls_table_1_4**, **bls_table_1_5**
- **bls_table_1_6**, **bls_table_1_8**, **bls_table_1_9**, **bls_table_1_10**, **bls_table_1_11**, **bls_table_1_12**

🔗 Next Steps for Tomorrow

Phase 0: File Organization & Cleanup (15-20 mins)

1. Move misplaced files to correct locations:

- Move `database-normalization-progress.md` from ChesapeakeTermite.com folder to lowrys-website project
- Review ChesapeakeTermite.com folder for any other lowrys-website files that need to be moved/deleted
- Clean up any duplicate or misplaced documentation files

2. Remove test files and temporary scripts:

- Delete test files (e.g., `page-test.tsx`, `page-backup.tsx`, `test-simple.tsx`)
- Remove temporary/unused scripts in the `scripts/` folder
- Clean up any backup files or old versions
- Remove unused API route test files

3. Update project documentation:

- Update `lowrys-website-structure.md` to reflect current file organization
- Document current API endpoints and their purposes
- Remove references to deleted/moved files
- Add documentation for new scripts and tools created

Phase 1: Table Analysis & Documentation (30 mins)

1. Analyze remaining specialized tables:

```
# Check structure and usage
SELECT COUNT(*) FROM occupation_data;
SELECT COUNT(*) FROM projections;
SELECT COUNT(*) FROM bls_special_tables;

# Get sample data
SELECT * FROM occupation_data LIMIT 5;
SELECT * FROM projections LIMIT 5;
SELECT * FROM bls_special_tables LIMIT 5;
```

2. Document table purposes:

- Create table usage documentation
- Identify which tables are actively used by API endpoints
- Determine if any can be consolidated or removed

Phase 2: BLS Table Normalization (1-2 hours)

3. Rename BLS tables for clarity:

```
-- Example renames
ALTER TABLE bls_table_1_1 RENAME TO bls_employment_matrix_major;
ALTER TABLE bls_table_1_2 RENAME TO bls_employment_matrix_minor;
-- etc.
```

4. Split year-based columns:

- Current format: `employment_2022`, `employment_2032`
- Target format: Separate rows with `year`, `employment` columns
- Create normalized structure: `(occupation_code, metric_type, year, value)`

5. Add metadata columns:

- `data_source` (workbook vs OEWS)
- `refresh_date`
- `import_timestamp`

Phase 3: Automation Script Updates (1 hour)

6. Update `scripts/import-bls-data.js`:

- Add year-based file detection
- Implement robust error handling
- Add metadata tracking for refresh dates
- Support both workbook and OEWS data sources

Phase 4: Validation & Testing (30 mins)

7. Validate normalization:

- Test all API endpoints still work
- Verify rate calculator functionality
- Check data integrity after normalization

Key Files & Locations

Scripts & Tools

- `scripts/cleanup-database.js` - Database cleanup automation
- `scripts/database-cleanup-backup.json` - Backup metadata
- `scripts/import-bls-data.js` - BLS data import automation (needs update)

Database Connection

- Environment: `.env.local`
- Database: `lowrys-bls-data` (Turso)
- Connection: Via `@libsql/client`

API Endpoints Using Database

- `/api/rate-calculator/route.ts` - Uses `occupations` table
 - `/api/seo-intake/route.ts` - Uses `seo_intake_forms` table
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Discovery Notes

PowerShell Compatibility

- Use semicolon (`;`) instead of `&&` for command chaining
- Turso CLI may not be globally available - Node.js scripts are more reliable

Database Access Pattern

- Direct SQL queries via `@libsql/client` work well
- Environment variables loaded from `.env.local`
- Backup/restore operations should preserve table schemas

BLS Data Structure Insights

- Tables 1-12 from BLS workbook (most empty, as noted previously)
 - Year-based column naming pattern across tables
 - Headers consistently on row 2 in source workbook
 - Both OEWS and workbook sources needed for complete data
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Quick Start Tomorrow

```
# Navigate to project
cd "c:\Users\jimlo\Desktop\Projects\lowrys-website"

# FIRST: Comprehensive file cleanup
# 1. Move database-normalization-progress.md from ChesapeakeTermite.com to lowrys-website
# 2. Review and delete test files (page-test.tsx, page-backup.tsx, test-simple.tsx, etc.)
# 3. Clean up scripts/ folder - remove temporary/unused scripts
# 4. Update lowrys-website-structure.md documentation
# 5. Remove any other backup/temp files

# Then run table analysis
node -e "
const { createClient } = require('@libsql/client');
require('dotenv').config({ path: '.env.local' });
const client = createClient({
  url: process.env.TURSO_DATABASE_URL,
  authToken: process.env.TURSO_AUTH_TOKEN,
});
// Add analysis queries here
"

# Continue with normalization tasks...
```

Status: Ready for comprehensive file cleanup + database analysis tomorrow! 🚀

CLEANUP PRIORITY:

1. **File Organization** - Move misplaced files, delete test files
2. **Documentation Update** - Update lowrys-website-structure.md
3. **Database Analysis** - Analyze table structure and usage
4. **Database Normalization** - Begin BLS table restructuring

NOTE: This progress file itself is in the wrong location (ChesapeakeTermite.com folder) and should be moved to the lowrys-website project folder first thing tomorrow.